Class_13

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Background

Today we will perform an RNASeq analysis on the effects of dexamethasone (hereafter "dex"), a common steroid, on airway smooth mucle (ASM) cell lines.

Data Import

We need two things for this analysis:

-**countData**: a table ith genes as rows and samples / experiments as columns.
-**colData**:metadata about the columns, (i.e. samples) in the main countData object.

```
counts <- read.csv("airway_scaledcounts.csv", row.names=1)
metadata <-  read.csv("airway_metadata.csv")
```

Now let's have a peek at the datasets:

```r
metadata
```

```
          id     dex celltype     geo_id
1 SRR1039508 control   N61311 GSM1275862
2 SRR1039509 treated   N61311 GSM1275863
3 SRR1039512 control  N052611 GSM1275866
4 SRR1039513 treated  N052611 GSM1275867
5 SRR1039516 control  N080611 GSM1275870
6 SRR1039517 treated  N080611 GSM1275871
7 SRR1039520 control  N061011 GSM1275874
8 SRR1039521 treated  N061011 GSM1275875
```

```r
head(counts)
```

```
                SRR1039508 SRR1039509 SRR1039512 SRR1039513 SRR1039516
ENSG00000000003        723        486        904        445       1170
ENSG00000000005          0          0          0          0          0
ENSG00000000419        467        523        616        371        582
ENSG00000000457        347        258        364        237        318
ENSG00000000460         96         81         73         66        118
ENSG00000000938          0          0          1          0          2
                SRR1039517 SRR1039520 SRR1039521
ENSG00000000003       1097        806        604
ENSG00000000005          0          0          0
ENSG00000000419        781        417        509
ENSG00000000457        447        330        324
ENSG00000000460         94        102         74
ENSG00000000938          0          0          0
```

Check on metadata counts correspondance

We need to check that the metadata matches the samples in our counts data.

```r
ncol(counts) == nrow(metadata)
```

```
[1] TRUE
```

```r
colnames(counts) == metadata$id
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many control samples are in this dataset?

```
sum(metadata$dex == "control")
```

```
[1] 4
```

Analysis Plan...

We have 4 replicates per condition ("control" and "treated"). We want to compare the control vs the treated to see which gene expression levels change when the drug is present.

We are going to analyze each row, gene by gene, to see the average values in the treatment columns and the average values in the control columns and see if they are different.

-Step 1. Find which columns in `counts` correspond to "control" samples.

-Step 2. Isolate these columns and calculate an average value fo each gene(i.e. each row)

-Step 3. Calculate an average value for each gene (i.e. each row).

```
# The indices (i.e. positions) that are "control"
control.inds<-metadata$dex == "control"
```

```
# Extract these "control" columns from counts
control.counts <- counts[,control.inds]
```

```
#Calculate the mean for each gene (i.e. row)
control.mean <- rowMeans(control.counts)
head(control.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
         900.75            0.00          520.50          339.75           97.25
ENSG00000000938
           0.75
```

Q. Do the same for the "treated" samples - find the mean count value per gene.

```r
# The indices (i.e. positions) that are "treated"
treated.inds<-metadata$dex == "treated"
```

```r
# Extract these "treated" columns from counts
treated.counts <- counts[,treated.inds]
```

```r
#Calculate the mean for each gene (i.e. row)
treated.mean <- rowMeans(treated.counts)
head(treated.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460
         658.00            0.00          546.00          316.50           78.75
ENSG00000000938
           0.00
```

Let's put these two mean values into a new data.fram `mean.counts` for book-keeping and plotting purposes.

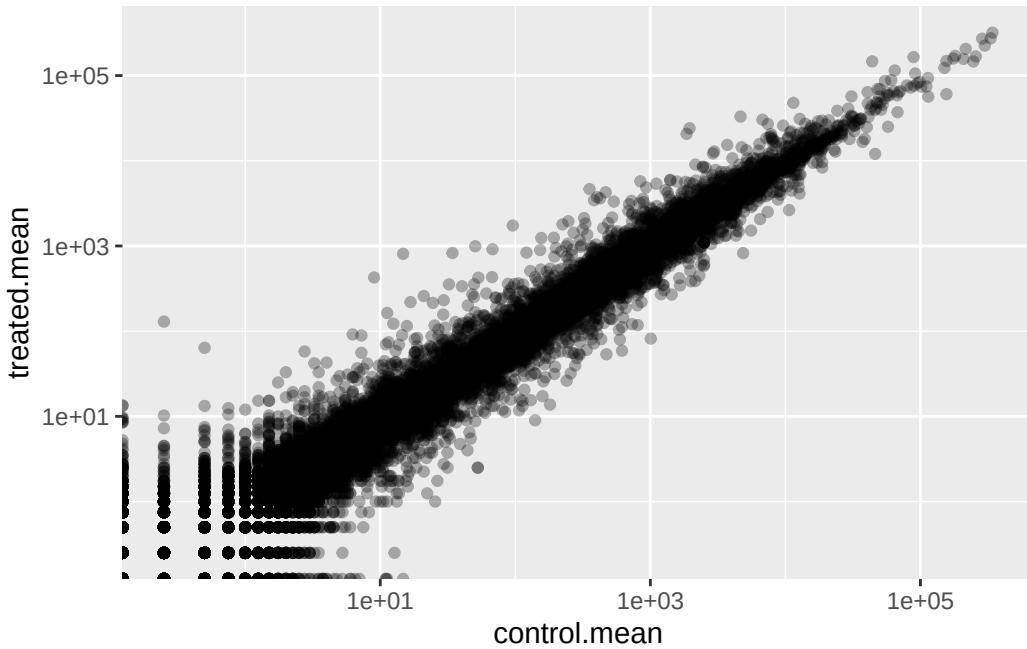
```r
meancounts <- data.frame(control.mean, treated.mean)
head(meancounts)
```

```
                control.mean treated.mean
ENSG00000000003       900.75       658.00
ENSG00000000005         0.00         0.00
ENSG00000000419       520.50       546.00
ENSG00000000457       339.75       316.50
ENSG00000000460        97.25        78.75
ENSG00000000938         0.75         0.00
```

Q. make a ggplot of average counts of control vs. treated.

```r
library(ggplot2)

ggplot(meancounts, aes(control.mean, treated.mean))+ geom_point(alpha=0.3) + scale_x_log10()
```

```
Warning in scale_x_log10(): log-10 transformation introduced infinite values.

Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```



Log2 units and fold change

If we consider "treated"/"control" counts we will get a number that tells us the change.

```
#No change
log2(20/20)
```

```
[1] 0
```

```
# A doubling in the treated vs control
log2(40/20)
```

```
[1] 1
```

```
# When treated is half the control
log2(10/20)
```

```
[1] -1
```

> Q. Add a new column called `log2fc` for log2 fold change of treated/control to our `meancounts` object.

```r
meancounts$log2fc <-  log2(meancounts$treated.mean/meancounts$control.mean)
head(meancounts)
```

```
                control.mean treated.mean      log2fc
ENSG00000000003       900.75       658.00 -0.45303916
ENSG00000000005         0.00         0.00         NaN
ENSG00000000419       520.50       546.00  0.06900279
ENSG00000000457       339.75       316.50 -0.10226805
ENSG00000000460        97.25        78.75 -0.30441833
ENSG00000000938         0.75         0.00        -Inf
```

Remove zero count genes

Typically we would not consider 0 count genes as they provide no data and should be excluded. These lead to "funky" log2 fold change values.

##DESeq analysis

We are missing any measure of significance from the work we have so far. Let's do this properly with the **DESeq2** package.

```r
library(DESeq2)
```

The DESeq2 package, like many bioconductor packages, wants it's input in a very specific way - a data structure setup with all the info it needs for the calculation.

```r
dds <- DESeqDataSetFromMatrix(countData = counts, colData = metadata, design = ~dex)
```

```
converting counts to integer mode
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

The main function in this package is called `DESeq()` it will run the full analysis for us on our dds' input object:

```r
dds <- DESeq(dds)
```

```
estimating size factors
```

```
estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

Extract our results:

```r
res <- results(dds)
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 6 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195     -0.3507030  0.168246 -2.084470 0.0371175
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160      0.2061078  0.101059  2.039475 0.0414026
ENSG00000000457 322.664844      0.0245269  0.145145  0.168982 0.8658106
ENSG00000000460  87.682625     -0.1471420  0.257007 -0.572521 0.5669691
ENSG00000000938   0.319167     -1.7322890  3.493601 -0.495846 0.6200029
                     padj
                <numeric>
ENSG00000000003  0.163035
ENSG00000000005        NA
ENSG00000000419  0.176032
ENSG00000000457  0.961694
ENSG00000000460  0.815849
ENSG00000000938        NA
```

```r
36000 * 0.05
```

```
[1] 1800
```

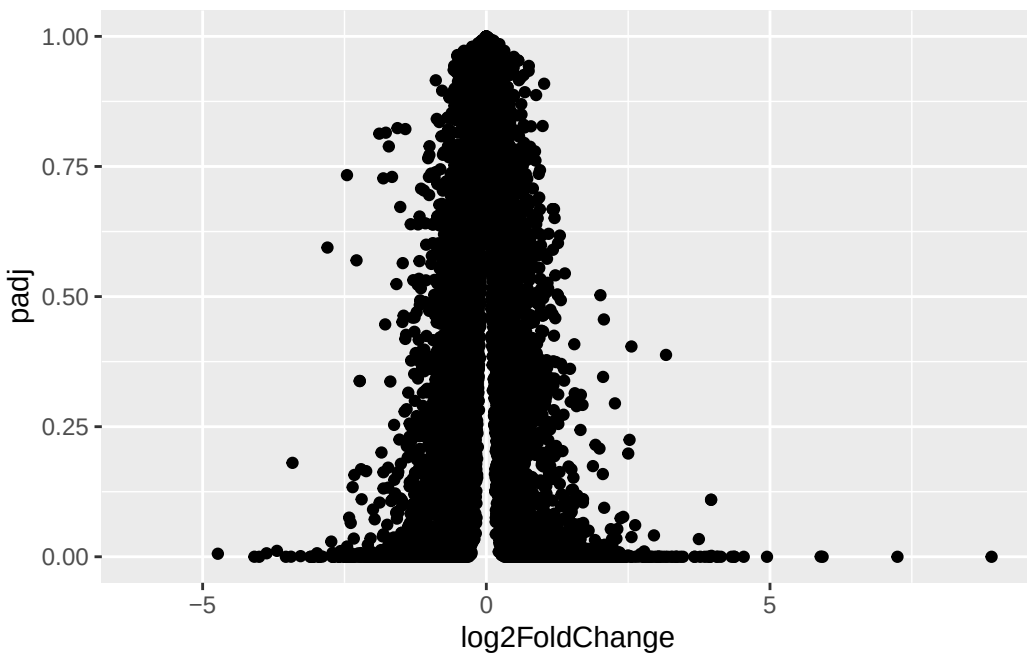
Volcano plot

A useful summary figure of our results is called a volcano plot. It is basically a plot of log2 fold change values vs Adjusted P-values.

Q.Use ggplot to make a first version "volcano plot" of `Log2FoldChange` vs `Padj`

```
ggplot(res , aes(log2FoldChange, padj))+ geom_point()
```

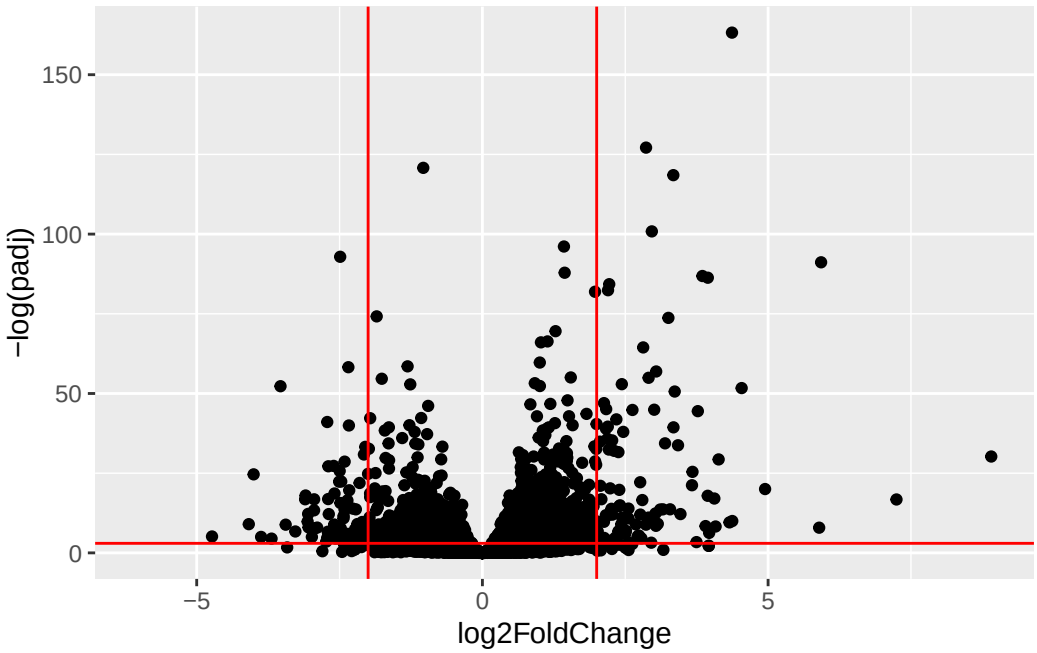
```
Warning: Removed 23549 rows containing missing values or values outside the scale range
(`geom_point()`).
```



This plot is not very useful since the y-axis, i.e the p-value, is not really helpful. Instead we want to focus on the low p-values.

```
ggplot(res , aes(log2FoldChange, -log(padj)))+ geom_point() + geom_vline(xintercept = c(-2,+2
```

```
Warning: Removed 23549 rows containing missing values or values outside the scale range
(`geom_point()`).
```



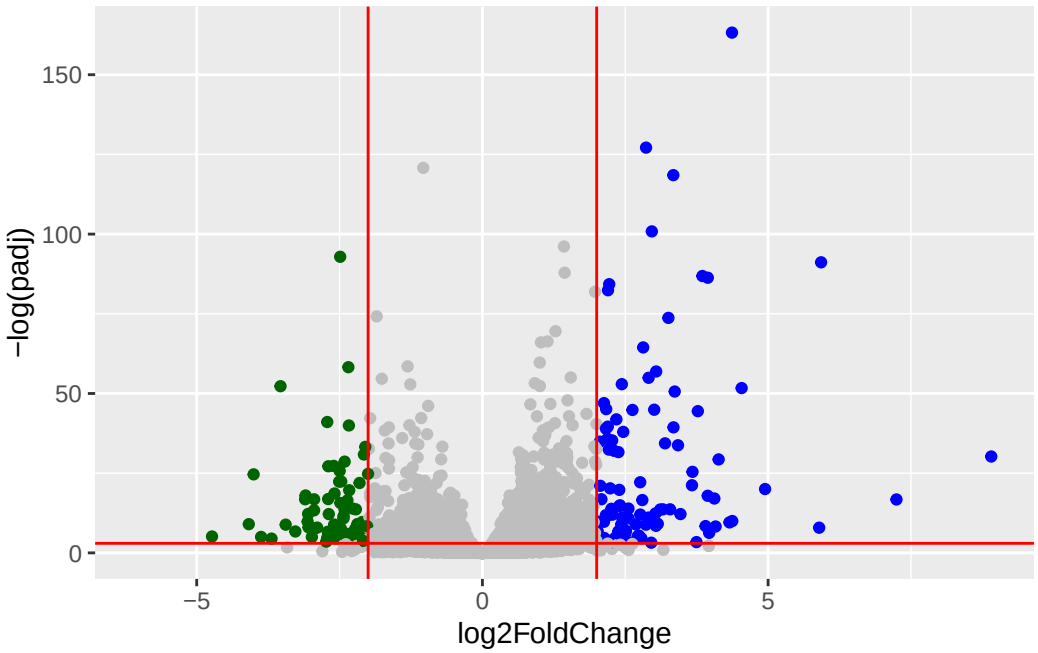
Add some plot annotation

> Q. Add color to the points (genes) we care about, nice axis lables, a useful title and a nice theme.

```
mycols <- rep("gray", nrow(res))
mycols[res$log2FoldChange > 2] <- "blue"
mycols[res$log2FoldChange < -2] <- "darkgreen"
mycols[res$padj >=0.05]<- "gray"
```

```
ggplot(res , aes(log2FoldChange, -log(padj)))+ geom_point(col = mycols) + geom_vline(xinterc
```

```
Warning: Removed 23549 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Save our results to a CSV file

```
write.csv(res, file="results.csv")
```

Add Annotation Data

To make sense of our results we need to know what the deferentially expressed genes are and what biological pathways and processes they are involved in.

Let's start by mapping our ENSEMBLE ids to the more conventional gene symbols.

We will use two bioconductor packages for this "mapping" "annotation" **AnnotationDbi** and **org.Hs.eg.db**.

We will first need to install these from biocunductor with `BiocManager::install("AnnotationDbi")`

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```r
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```r
res$symbol <- mapIds(org.Hs.eg.db,
       keys = rownames(res),
       keytype= "ENSEMBL",
       column="SYMBOL")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 7 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195     -0.3507030  0.168246 -2.084470 0.0371175
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160      0.2061078  0.101059  2.039475 0.0414026
ENSG00000000457 322.664844      0.0245269  0.145145  0.168982 0.8658106
ENSG00000000460  87.682625     -0.1471420  0.257007 -0.572521 0.5669691
ENSG00000000938   0.319167     -1.7322890  3.493601 -0.495846 0.6200029
                     padj      symbol
                <numeric> <character>
ENSG00000000003  0.163035      TSPAN6
ENSG00000000005        NA        TNMD
ENSG00000000419  0.176032        DPM1
ENSG00000000457  0.961694       SCYL3
ENSG00000000460  0.815849       FIRRM
ENSG00000000938        NA         FGR
```

Q. Can you add "GENENAME" an "ENTREZID" as new columns to `res` named "name" and "entrez"?

```r
res$name <- mapIds(org.Hs.eg.db,
       keys = rownames(res),
       keytype= "ENSEMBL",
       column="GENENAME")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez <- mapIds(org.Hs.eg.db,
       keys = rownames(res),
       keytype= "ENSEMBL",
       column="ENTREZID")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 9 columns
                  baseMean log2FoldChange     lfcSE      stat    pvalue
                 <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000000003 747.194195     -0.3507030  0.168246 -2.084470 0.0371175
ENSG00000000005   0.000000             NA        NA        NA        NA
ENSG00000000419 520.134160      0.2061078  0.101059  2.039475 0.0414026
ENSG00000000457 322.664844      0.0245269  0.145145  0.168982 0.8658106
ENSG00000000460  87.682625     -0.1471420  0.257007 -0.572521 0.5669691
ENSG00000000938   0.319167     -1.7322890  3.493601 -0.495846 0.6200029
                     padj      symbol                   name      entrez
                <numeric> <character>            <character> <character>
ENSG00000000003  0.163035      TSPAN6          tetraspanin 6        7105
ENSG00000000005        NA        TNMD            tenomodulin       64102
ENSG00000000419  0.176032        DPM1 dolichyl-phosphate m..        8813
ENSG00000000457  0.961694       SCYL3 SCY1 like pseudokina..       57147
ENSG00000000460  0.815849       FIRRM FIGNL1 interacting r..       55732
ENSG00000000938        NA         FGR FGR proto-oncogene, ..        2268
```

```r
write.csv(res, file="results_annotated.csv")
```

Pathway analysis

Now we know the gene naes (gene symbols) and their entrez IDs we can find out what pathways they are involved in. Thi is called "pathway analysis" or "gene set enrichment" .

We will use the **gage** package and the **pathviewer** package to do this analysis (but there are loads of others).

```
library(pathview)
library(gage)
library(gageData)
```

Let's see what is in gageData, specifically KEGG pathways.

```
data(kegg.sets.hs)
head(kegg.sets.hs, 2)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"
```

To run our pathway analysis we will use the **gage()** function. It wants two main inputs: a vector of importance (in our case the log2 fold change values); and the gene sets to check overlap for.

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$symbol
head(foldchanges)
```

```
     TSPAN6        TNMD        DPM1       SCYL3       FIRRM         FGR
-0.35070302          NA  0.20610777  0.02452695 -0.14714205 -1.73228897
```

KEGG speaks entrez(i.e uses ENTREZID format) not gene symbol format.

```
names(foldchanges) <-  res$entrez
```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

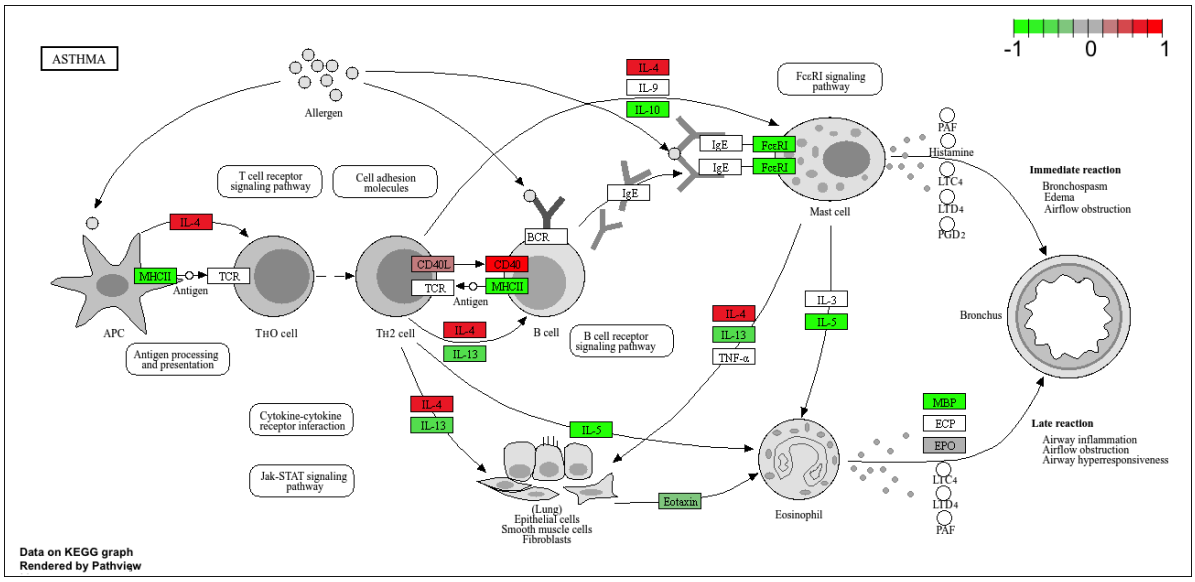
```
head(keggres$less, 5)
```

```
                                                      p.geomean stat.mean
hsa05332 Graft-versus-host disease                   0.0004250461 -3.473346
hsa04940 Type I diabetes mellitus                    0.0017820293 -3.002352
hsa05310 Asthma                                      0.0020045888 -3.009050
hsa04672 Intestinal immune network for IgA production 0.0060434515 -2.560547
hsa05330 Allograft rejection                         0.0073678825 -2.501419
                                                             p.val      q.val
hsa05332 Graft-versus-host disease                   0.0004250461 0.09053483
hsa04940 Type I diabetes mellitus                    0.0017820293 0.14232581
hsa05310 Asthma                                      0.0020045888 0.14232581
hsa04672 Intestinal immune network for IgA production 0.0060434515 0.31387180
hsa05330 Allograft rejection                         0.0073678825 0.31387180
                                                     set.size         exp1
hsa05332 Graft-versus-host disease                         40 0.0004250461
hsa04940 Type I diabetes mellitus                          42 0.0017820293
hsa05310 Asthma                                            29 0.0020045888
hsa04672 Intestinal immune network for IgA production      47 0.0060434515
hsa05330 Allograft rejection                               36 0.0073678825
```

Let's make a figure of one of these pathways with our DEGs highlighted:

```
pathview(foldchanges, pathway.id = "hsa05310")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class13

Info: Writing image file hsa05310.pathview.png
```



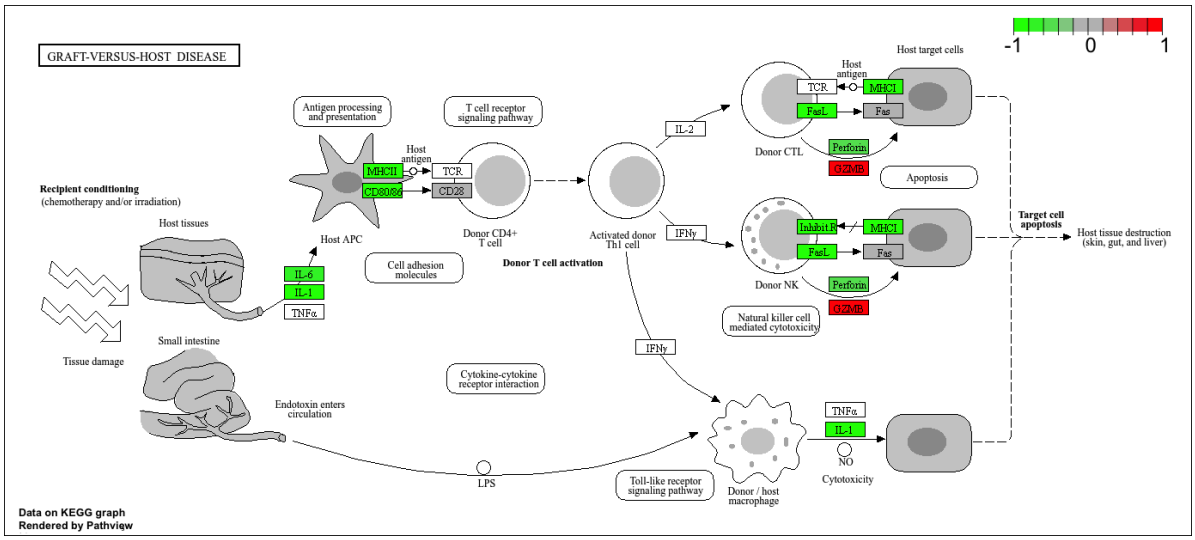
Q. Generate and insert a pathway figure for "Graft-versushost disease" and "Type I diabetes".

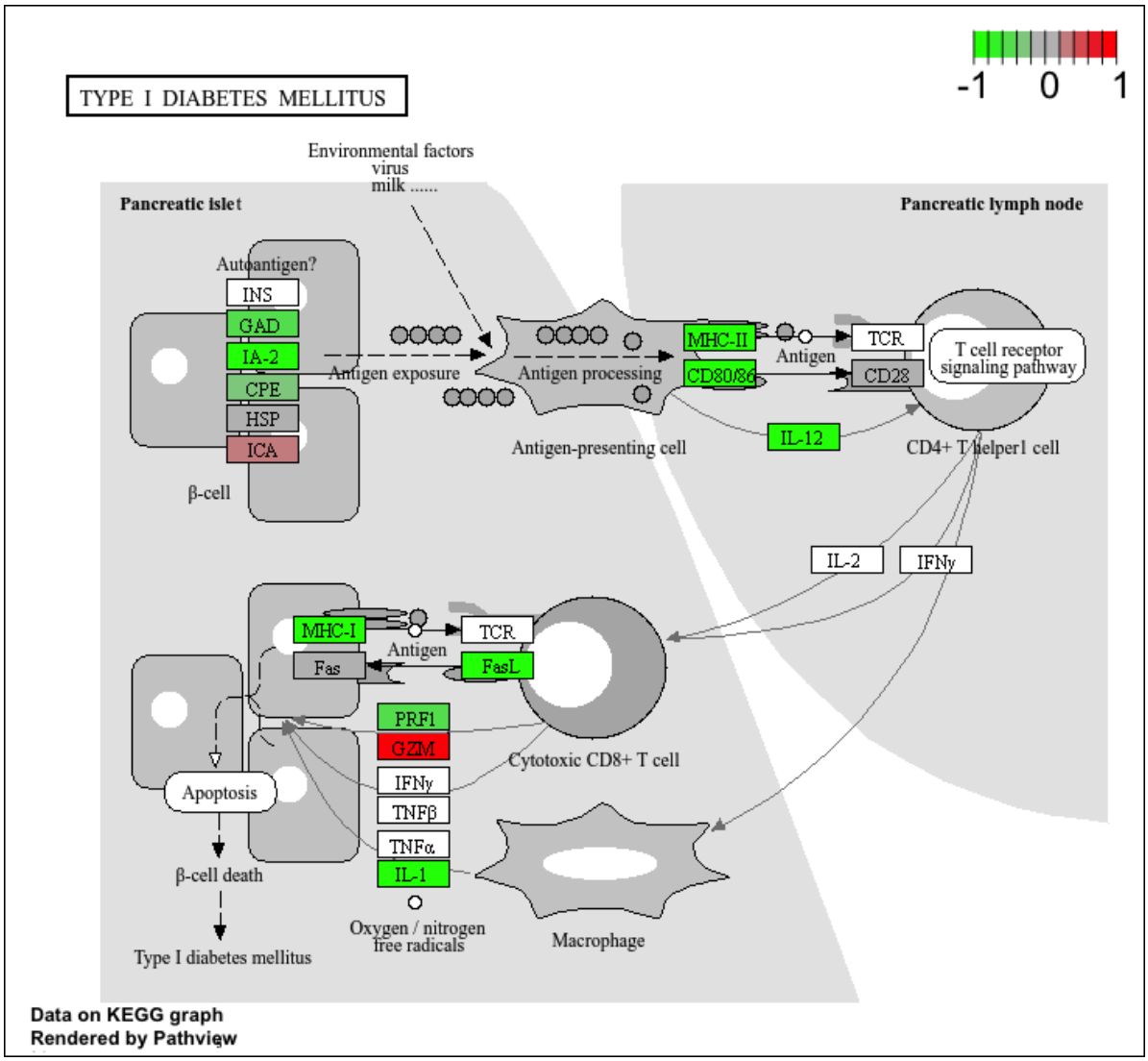
```
pathview(foldchanges, pathway.id = "hsa05332")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class13

Info: Writing image file hsa05332.pathview.png
```

```
pathview(foldchanges, pathway.id = "hsa04940")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class13

Info: Writing image file hsa04940.pathview.png
```


GRAFT-VERSUS-HOST DISEASE
-1
0
1
Host target cells
Host antigen
TCR
MHCI
FasL
Fas
Donor CTL
Perforin
GZMB
Apoptosis
Antigen processing and presentation
T cell receptor signaling pathway
Host antigen
MHCII
CD80/86
TCR
CD28
IL-2
Recipient conditioning
(chemotherapy and/or irradiation)
Host tissues
Host APC
Donor CD4+ T cell
Activated donor Th1 cell
IFNγ
Inhibit R
MHCI
FasL
Fas
Donor NK
Perforin
GZMB
Target cell apoptosis
Host tissue destruction (skin, gut, and liver)
IL-6
IL-1
TNFα
Cell adhesion molecules
Donor T cell activation
Natural killer cell mediated cytotoxicity
Tissue damage
Small intestine
IFNγ
Cytokine-cytokine receptor interaction
Endotoxin enters circulation
TNFα
IL-1
NO
Cytotoxicity
LPS
Toll-like receptor signaling pathway
Donor / host macrophage
Data on KEGG graph
Rendered by Pathview



Data on KEGG graph
Rendered by Pathview